FACULTY OF COMPUTING & INFORMATICS (FCI)

CSN6224
COMPUTER NETWORKS

ASSIGNMENT

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GROUP 20

TC4L

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Introduction

This report presents the design, configuration, and simulation of a **high-availability dual-stack IPv4 and IPv6 network** using **Cisco Packet Tracer**, developed as part of the **CSN6224**

Computer Networks assignment. The objective of this project is to demonstrate practical understanding of modern network technologies, including IP addressing, VLAN segmentation, dynamic routing, server deployment, wireless networking, and redundancy mechanisms.

The proposed network consists of multiple interconnected networks, each serving a specific purpose such as IPv4 user access, IPv6 user access, dual-stack server services, and secure wireless connectivity. To ensure scalability and resilience, the network incorporates **VLANs**, **trunk links**, **Spanning Tree Protocol (STP)**, and **dynamic routing protocols** for both IPv4 and IPv6. Core services such as **DNS, HTTP, FTP, and Email** are deployed within a dedicated dual-stack server network to support end-to-end connectivity across all segments.

In addition, a secure wireless network is implemented using **RADIUS-based authentication**, enabling controlled access for wireless users while maintaining network security. Comprehensive connectivity testing and simulation mode verification are conducted to validate correct routing, service availability, and protocol operation. Overall, this project demonstrates the practical application of theoretical networking concepts in designing a reliable and scalable enterprise-style network.

2. IP addressing plan (IPv4 & IPv6)

2.1 Pink Network – IPv4 User Network

The Pink Network is an IPv4 user network configured using DHCP. Router2 acts as the default gateway for the subnet 202.2.5.0/24. End-user devices dynamically obtain IP addresses from the DHCP server, while the DHCP server itself is configured with a static IPv4 address to ensure consistent service availability.

Device	**VLAN**	**IP Address**	**Subnet Mask**	**Default Gateway**
Router2	VLAN10	202.2.5.1	255.255.255.0	–
PC0	VLAN10	DHCP (202.2.5.4)	255.255.255.0	202.2.5.1

PC1	VLAN10	DHCP	255.255.255.0	202.2.5.1
DHCP Server	VLAN10	202.2.5.10	255.255.255.0	202.2.5.1

2.2 Purple Network – Dual-Stack Server Network

The Purple Network hosts the organisation's core services and is configured as a dual-stack network supporting both IPv4 and IPv6. All servers are assigned static IPv4 and IPv6 addresses to ensure service stability. Router1 acts as the default gateway for both protocols, enabling full connectivity between user, wireless, and server networks

Device	VLAN	IPv4 Address	Subnet Mask	IPv4 Gateway	IPv6 Address	IPv6 Gateway
Router 1	VLAN Server	202.11.6.1	255.255.255.0	–	2001:DB8:1106:2004::1	–
DNS Server	VLAN Server	202.11.6.10	255.255.255.0	202.11.6.1	2001:DB8:1106:2004::10	2001:DB8:1106:2004::1
Web Server	VLAN Server	202.11.6.11	255.255.255.0	202.11.6.1	2001:DB8:1106:2004::11	2001:DB8:1106:2004::1
Email Server	VLAN Server	202.11.6.12	255.255.255.0	202.11.6.1	2001:DB8:1106:2004::12	2001:DB8:1106:2004::1
FTP Server	VLAN Server	202.11.6.13	255.255.255.0	202.11.6.1	2001:DB8:1106:2004::13	2001:DB8:1106:2004::1

2.3 Blue Network – IPv6 User Network

The Blue Network is configured as an IPv6-only user network. IPv6 addresses are automatically assigned to client devices using SLAAC, while Router3 serves as the IPv6 default gateway. DNS resolution is provided by the dual-stack DNS server located in the Purple network, enabling IPv6 clients to access all services.

Device	Network	IPv6 Address	Prefix	IPv6 Gateway	DNS Server
Router3	Blue	2001:DB8:2006:2002::1	/64	–	–
PC2	Blue	2001:DB8:2006:2002:3543:B592:5206:1ABD	/64	FE80::210:11FF:FE05:5601	2001:DB8:1106:2004::10

2.4 Wireless Network – IPv4 Wireless Access

The wireless network is configured using IPv4 addressing with DHCP. Router4 acts as the default gateway and provides inter-VLAN routing for the wireless VLANs. Wireless clients successfully obtain IP addresses dynamically and are able to access internal servers and services through the core network.

Device	VLAN	IP Address	Subnet Mask	Default Gateway
Router4	VLAN Wireless	202.21.6.1	255.255.255.0	–
Laptop3	VLAN Wireless	DHCP (202.21.6.22)	255.255.255.0	202.21.6.1

3.0 SUBNETTING TABLES

All IPv4 networks in this project are allocated using /24 subnet masks, as specified in the assignment guidelines and derived from group members' dates of birth.

Pink Network – IPv4 User Network

Network: 202.2.5.0/24

Item	Value
Network Address	202.2.5.0
Subnet Mask	255.255.255.0
CIDR Notation	/24
Usable IP Range	202.2.5.1 – 202.2.5.254
Broadcast Address	202.2.5.255
Default Gateway	202.2.5.1
Total Usable Hosts	254

Purple Network – Dual-Stack Server Network

Network: 202.11.6.0/24

Item	Value
Network Address	202.11.6.0
Subnet Mask	255.255.255.0
CIDR Notation	/24
Usable IP Range	202.11.6.1 – 202.11.6.254
Broadcast Address	202.11.6.255
Default Gateway	202.11.6.1
Total Usable Hosts	254

3.3 Wireless Network – IPv4 Wireless LAN

Network: 202.21.6.0/24

Item	Value
Network Address	202.21.6.0
Subnet Mask	255.255.255.0
CIDR Notation	/24

Usable IP Range	202.21.6.1 – 202.21.6.254
Broadcast Address	202.21.6.255
Default Gateway	202.21.6.1
Total Usable Hosts	254

3.4 Wireless VLAN Subnets (Router4)

VLAN 30 – WLAN 1

Network: 202.21.30.0/24

Item	Value
Network Address	202.21.30.0
Subnet Mask	255.255.255.0
CIDR	/24
Usable IP Range	202.21.30.1 – 202.21.30.254
Broadcast	202.21.30.255
Gateway	202.21.30.1

VLAN 40 – WLAN 2

Network: 202.21.40.0/24

Item	Value
Network Address	202.21.40.0
Subnet Mask	255.255.255.0
CIDR	/24
Usable IP Range	202.21.40.1 – 202.21.40.254
Broadcast	202.21.40.255
Gateway	202.21.40.1

VLAN 50 – WLAN 3

Network: 202.21.50.0/24

Item	Value
Network Address	202.21.50.0
Subnet Mask	255.255.255.0
CIDR	/24
Usable IP Range	202.21.50.1 – 202.21.50.254
Broadcast	202.21.50.255
Gateway	202.21.50.1

2.5 IPv6 Subnetting (Blue Network)

IPv6 Prefix: `2001:DB8:2006:2002::/64`

Item	Value
Network Prefix	2001:DB8:2006:2002::
Prefix Length	/64
Addressing Method	SLAAC
Default Gateway	2001:DB8:2006:2002::1
Address Space	2^{64} addresses

2.6 IPv6 Subnetting (Purple Dual-Stack Network)

IPv6 Prefix: `2001:DB8:1106:2004::/64`

Item	Value
Network Prefix	2001:DB8:1106:2004::
Prefix Length	/64

Default Gateway	2001:DB8:1106:2004::1
Address Space	2^{64} addresses

4.1 VLAN Segmentation & Switch Configuration

VLANs were configured on access and core switches to logically separate user, server, IPv6, and wireless networks. Trunk links were enabled between switches to allow VLAN traffic to traverse the core network. Spanning Tree Protocol (STP) was used to prevent switching loops and ensure high availability.

```
Switch>show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/1, Gig0/2
10   Student                          active
20   Staff                            active
30   Server                           active
40   Wireless                         active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch>
```

- ```
  Switch>show interfaces trunk
  Port        Mode         Encapsulation  Status        Native vlan
  Fa0/1       on           802.1q         trunking      1
  Fa0/2       on           802.1q         trunking      1
  Fa0/3       on           802.1q         trunking      1

  Port        Vlans allowed on trunk
  Fa0/1       1-1005
  Fa0/2       1-1005
  Fa0/3       1-1005

  Port        Vlans allowed and active in management domain
  Fa0/1       1,10,20,30,40
  Fa0/2       1,10,20,30,40
  Fa0/3       1,10,20,30,40

  Port        Vlans in spanning tree forwarding state and not pruned
  Fa0/1       1,10,20,30,40
  Fa0/2       1,10,20,30,40
  Fa0/3       1,10,20,30,40
  ```
  ```

```
Switch>show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0001.C70C.5C63
             Cost        19
             Port        2(FastEthernet0/2)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     00D0.FF7A.EE63
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/1            Desg FWD 19        128.1    P2p
Fa0/2            Root FWD 19        128.2    P2p
Fa0/3            Desg FWD 19        128.3    P2p

VLAN0010
  Spanning tree enabled protocol ieee
  Root ID    Priority    32778
             Address     0001.C70C.5C63
             Cost        19
             Port        2(FastEthernet0/2)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32778  (priority 32768 sys-id-ext 10)
             Address     00D0.FF7A.EE63
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/1            Desg FWD 19        128.1    P2p
Fa0/2            Root FWD 19        128.2    P2p
Fa0/3            Desg FWD 19        128.3    P2p

VLAN0020
  Spanning tree enabled protocol ieee
  Root ID    Priority    32788
             Address     0001.C70C.5C63
             Cost        19
             Port        2(FastEthernet0/2)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32788  (priority 32768 sys-id-ext 20)
             Address     00D0.FF7A.EE63
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
```

```
Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/1            Desg FWD 19        128.1    P2p
Fa0/2            Root FWD 19        128.2    P2p
Fa0/3            Desg FWD 19        128.3    P2p

VLAN0030
  Spanning tree enabled protocol ieee
  Root ID    Priority    32798
             Address     0001.C70C.5C63
             Cost        19
             Port        2(FastEthernet0/2)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32798  (priority 32768 sys-id-ext 30)
             Address     00D0.FF7A.EE63
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/1            Desg FWD 19        128.1    P2p
Fa0/2            Root FWD 19        128.2    P2p
Fa0/3            Desg FWD 19        128.3    P2p

VLAN0040
  Spanning tree enabled protocol ieee
  Root ID    Priority    32808
             Address     0001.C70C.5C63
             Cost        19
             Port        2(FastEthernet0/2)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32808  (priority 32768 sys-id-ext 40)
             Address     00D0.FF7A.EE63
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/1            Desg FWD 19        128.1    P2p
Fa0/2            Root FWD 19        128.2    P2p
Fa0/3            Desg FWD 19        128.3    P2p
```

4.2 IPv4 & IPv6 Address Configuration

IPv4 and IPv6 addressing was configured on routers and end devices according to the assigned addressing plan. IPv4 addresses were either statically assigned or dynamically allocated via DHCP, while IPv6 addresses were configured using a combination of static addressing and SLAAC. This ensured full dual-stack connectivity across the network.

IPv4 Router Proof

```
Router>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     202.2.5.1       YES manual up                    up
GigabitEthernet0/1     10.0.0.2        YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router>
```

Figure 4.4: IPv4 interface configuration on Router2

IPv6 Router Proof

```
Router>show ipv6 interface brief
GigabitEthernet0/0         [up/up]
    FE80::210:11FF:FE05:5601
GigabitEthernet0/1         [up/up]
    FE80::210:11FF:FE05:5602
    2001:DB8:2006:2002::1
GigabitEthernet0/2         [administratively down/down]
    unassigned
Vlan1                      [administratively down/down]
    unassigned
Router>
```

Figure 4.5: IPv6 interface configuration on Router3

Dual-Stack Router Proof

```
Router>
Router>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     202.11.6.1      YES manual up                    up
GigabitEthernet0/1     10.0.0.1        YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router>
```

Figure 4.6: Dual-stack IPv4 configuration on Router1

4.3 Dual-Stack Server Configuration

The server network was configured as a dual-stack environment to support both IPv4 and IPv6 communication. Core services including DNS, HTTP, Email, and FTP were deployed with static addressing to ensure reliability and continuous availability for all user and wireless networks.

DNS Server

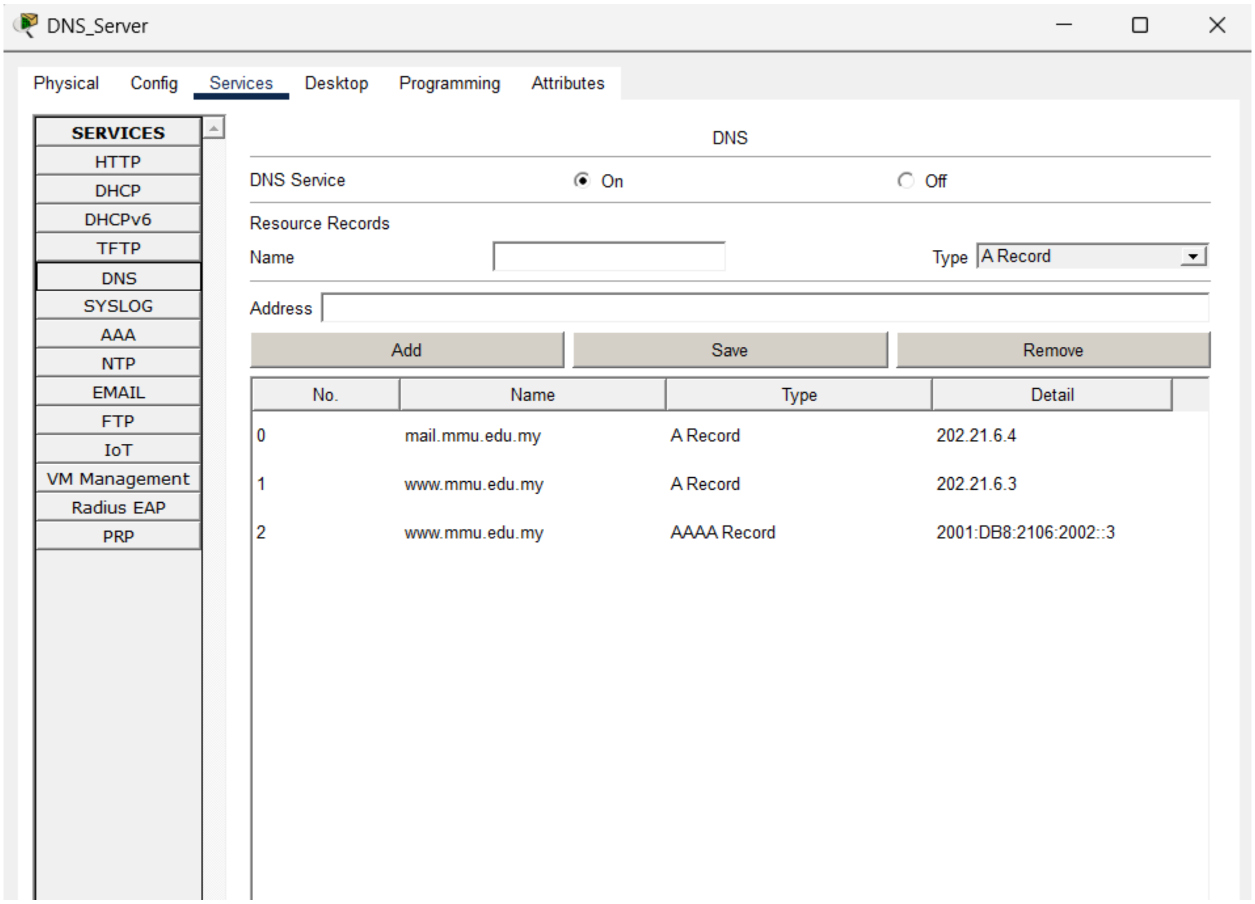


Figure 4.7: DNS service configuration on dual-stack DNS server

Web Server

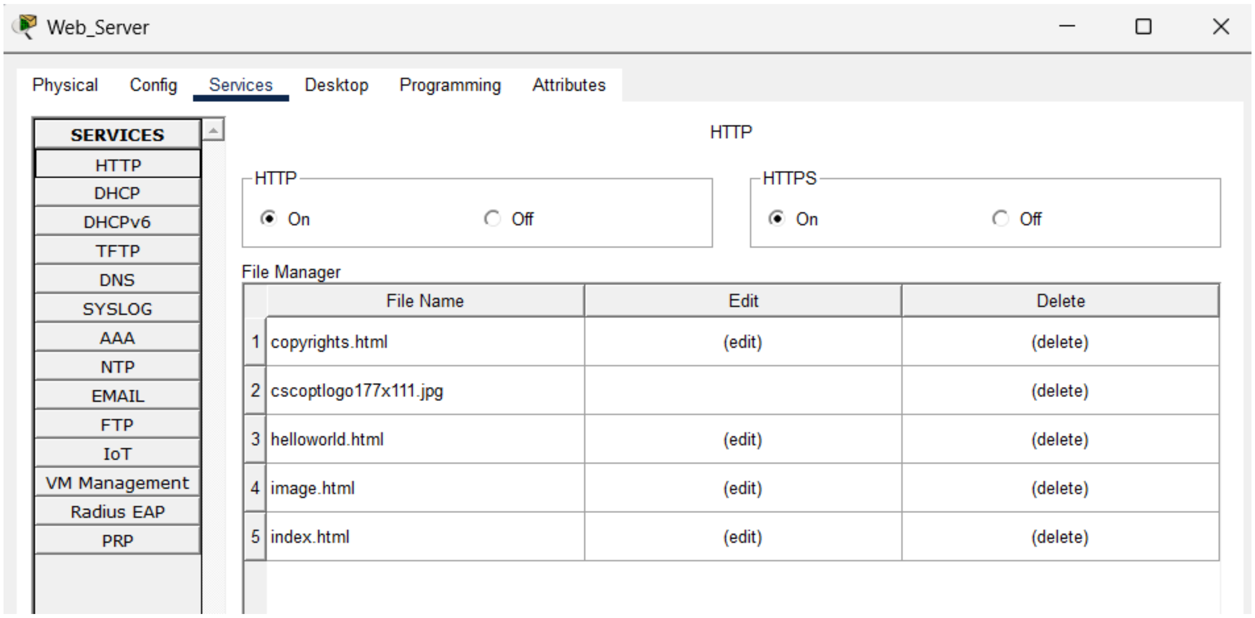


Figure 4.8: HTTP web service configuration

Email Server

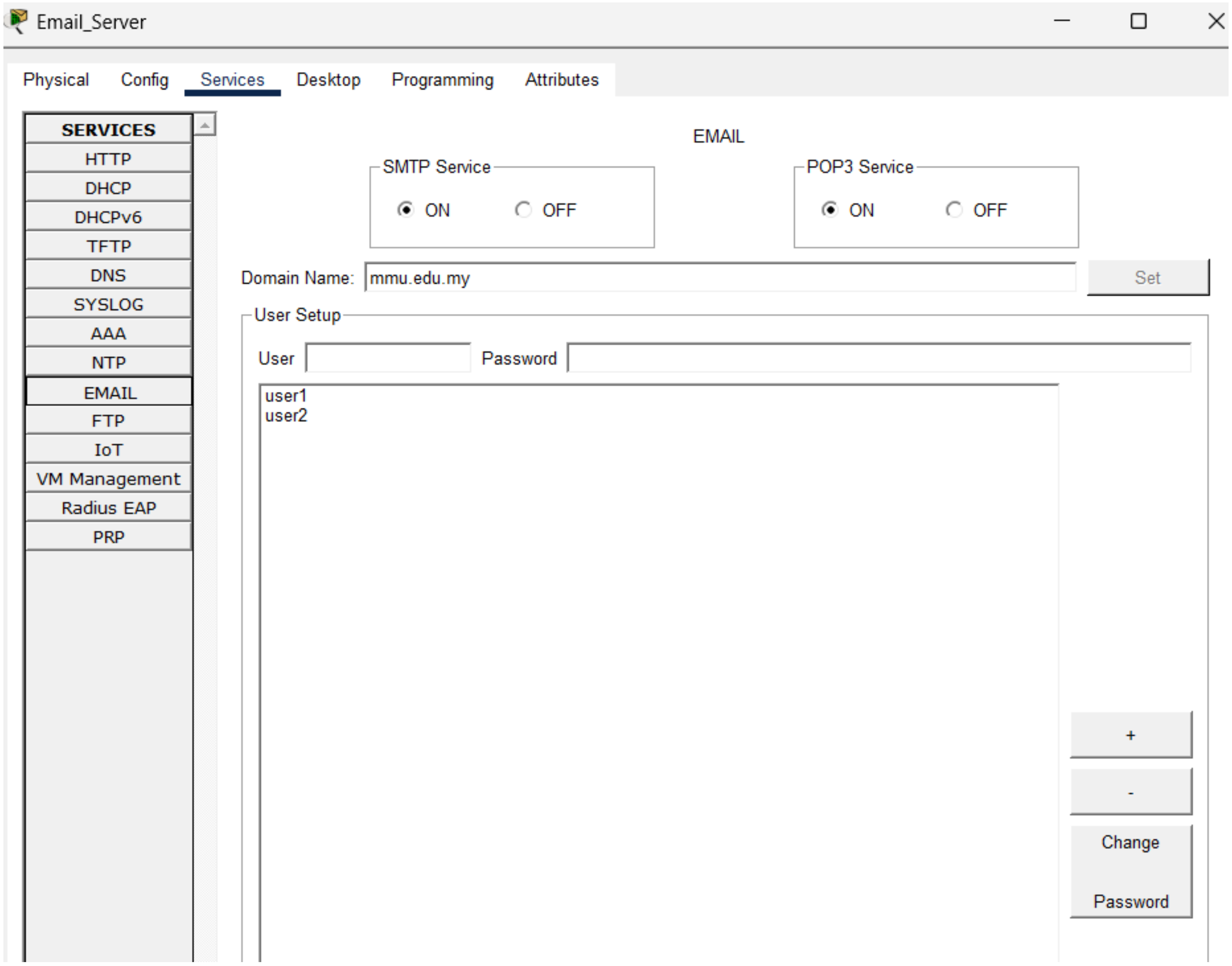


Figure 4.9: Email service configuration

FTP Server

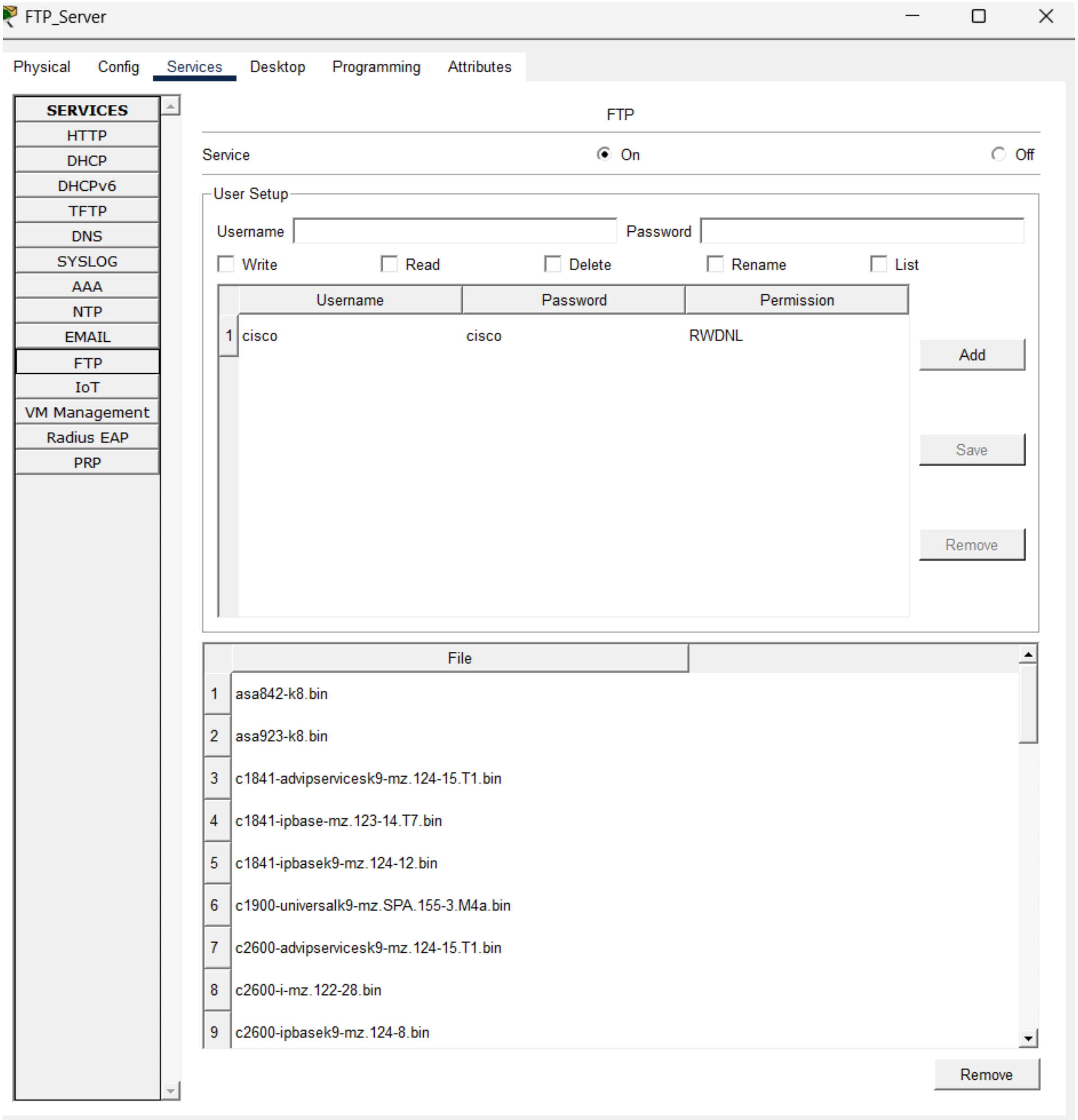


Figure 4.10: FTP service configuration

4.4 Dynamic Routing (IPv4 + IPv6)

Dynamic routing was implemented to enable automatic route exchange between all networks. OSPFv2 was used for IPv4 routing, while OSPFv3 was configured for IPv6 routing. This allows scalable and resilient communication between user, server, and wireless networks without manual static routes.

IPv4 routing table

```
Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/24 is directly connected, GigabitEthernet0/1
L       10.0.0.3/32 is directly connected, GigabitEthernet0/1
O    202.2.5.0/24 [110/2] via 10.0.0.2, 00:44:29, GigabitEthernet0/1
O    202.11.6.0/24 [110/2] via 10.0.0.1, 00:44:29, GigabitEthernet0/1
     202.20.6.0/24 is variably subnetted, 2 subnets, 2 masks
C       202.20.6.0/24 is directly connected, GigabitEthernet0/0
L       202.20.6.1/32 is directly connected, GigabitEthernet0/0
```

Figure 4.11: IPv4 routing table showing dynamically learned routes

IPv6 routing table

```
Router>show ipv6 route
IPv6 Routing Table - 3 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       D - EIGRP, EX - EIGRP external
C   2001:DB8:2006:2002::/64 [0/0]
     via GigabitEthernet0/1, directly connected
L   2001:DB8:2006:2002::1/128 [0/0]
     via GigabitEthernet0/1, receive
L   FF00::/8 [0/0]
     via Null0, receive
Router>
```

Figure 4.12: IPv6 routing table showing OSPFv3 learned routes

4.5 Wireless + RADIUS Configuration

A wireless network was deployed using Router4 with multiple VLANs to support different user groups. RADIUS authentication was implemented to provide secure access control. Wireless clients authenticate successfully and obtain IP addresses dynamically via DHCP.

RADIUS Server

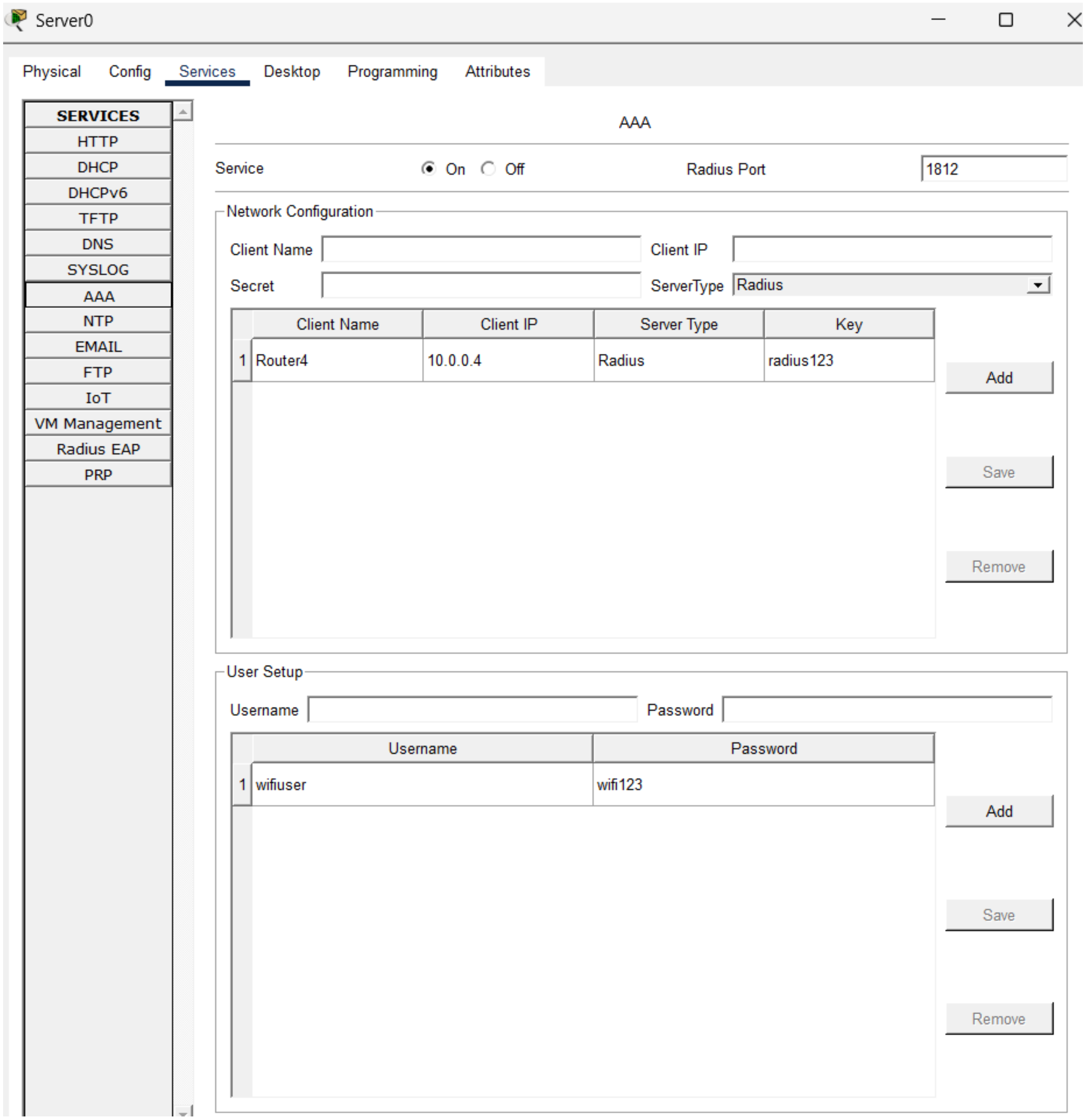


Figure 4.13: RADIUS server configuration for wireless authentication

Router4 AAA config

```
Router>enable
Router#show running-config | section aaa
aaa new-model
aaa authentication login WIFI_AUTH group radius local
aaa authorization network default group radius
Router#
```

Figure 4.14: AAA configuration on Router4

Wireless client connected

Figure 4.15: Successful wireless authentication via RADIUS

5.1 Connectivity Tests

5.1.1 Overview

Connectivity tests were conducted to verify end-to-end communication across the implemented IPv4 and IPv6 networks. The tests include ICMP reachability, application-layer service access, and wireless authentication to ensure that routing, VLAN segmentation, and dual-stack configurations function correctly.

TEST 1 — IPv4 Connectivity (Wired → Server)

```
C:\>ping 202.11.6.10

Pinging 202.11.6.10 with 32 bytes of data:

Reply from 202.11.6.10: bytes=32 time<1ms TTL=126
Reply from 202.11.6.10: bytes=32 time<1ms TTL=126
Reply from 202.11.6.10: bytes=32 time<1ms TTL=126
Reply from 202.11.6.10: bytes=32 time<1ms TTL=126

Ping statistics for 202.11.6.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Figure 5.1: Successful IPv4 ping from wired client to DNS server

TEST 2 — IPv4 Connectivity (Wireless → Server)

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 202.11.6.11

Pinging 202.11.6.11 with 32 bytes of data:

Request timed out.
Reply from 202.11.6.11: bytes=32 time=6ms TTL=126
Reply from 202.11.6.11: bytes=32 time=8ms TTL=126
Reply from 202.11.6.11: bytes=32 time=12ms TTL=126

Ping statistics for 202.11.6.11:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 6ms, Maximum = 12ms, Average = 8ms

C:\>
```

Figure 5.2: IPv4 connectivity test from wireless client to web server

TEST 3 — IPv6 Connectivity (Client → Gateway, Link-Local)

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping FE80::210:11FF:FE05:5601

Pinging FE80::210:11FF:FE05:5601 with 32 bytes of data:

Reply from FE80::210:11FF:FE05:5601: bytes=32 time<1ms TTL=255
Reply from FE80::210:11FF:FE05:5601: bytes=32 time<1ms TTL=255
Reply from FE80::210:11FF:FE05:5601: bytes=32 time<1ms TTL=255
Reply from FE80::210:11FF:FE05:5601: bytes=32 time<1ms TTL=255

Ping statistics for FE80::210:11FF:FE05:5601:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Figure 5.3: IPv6 link-local connectivity between Blue network client and router gateway

TEST 4 — Web Server Access (HTTP)

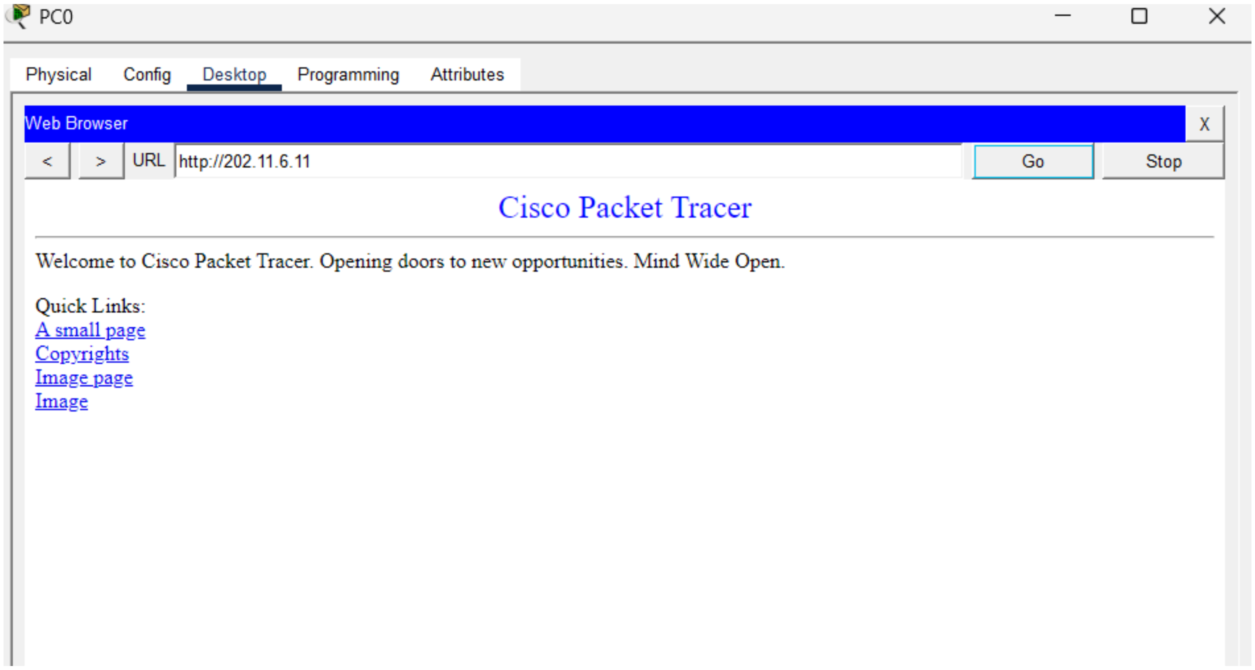


Figure 5.4: Successful HTTP access to web server

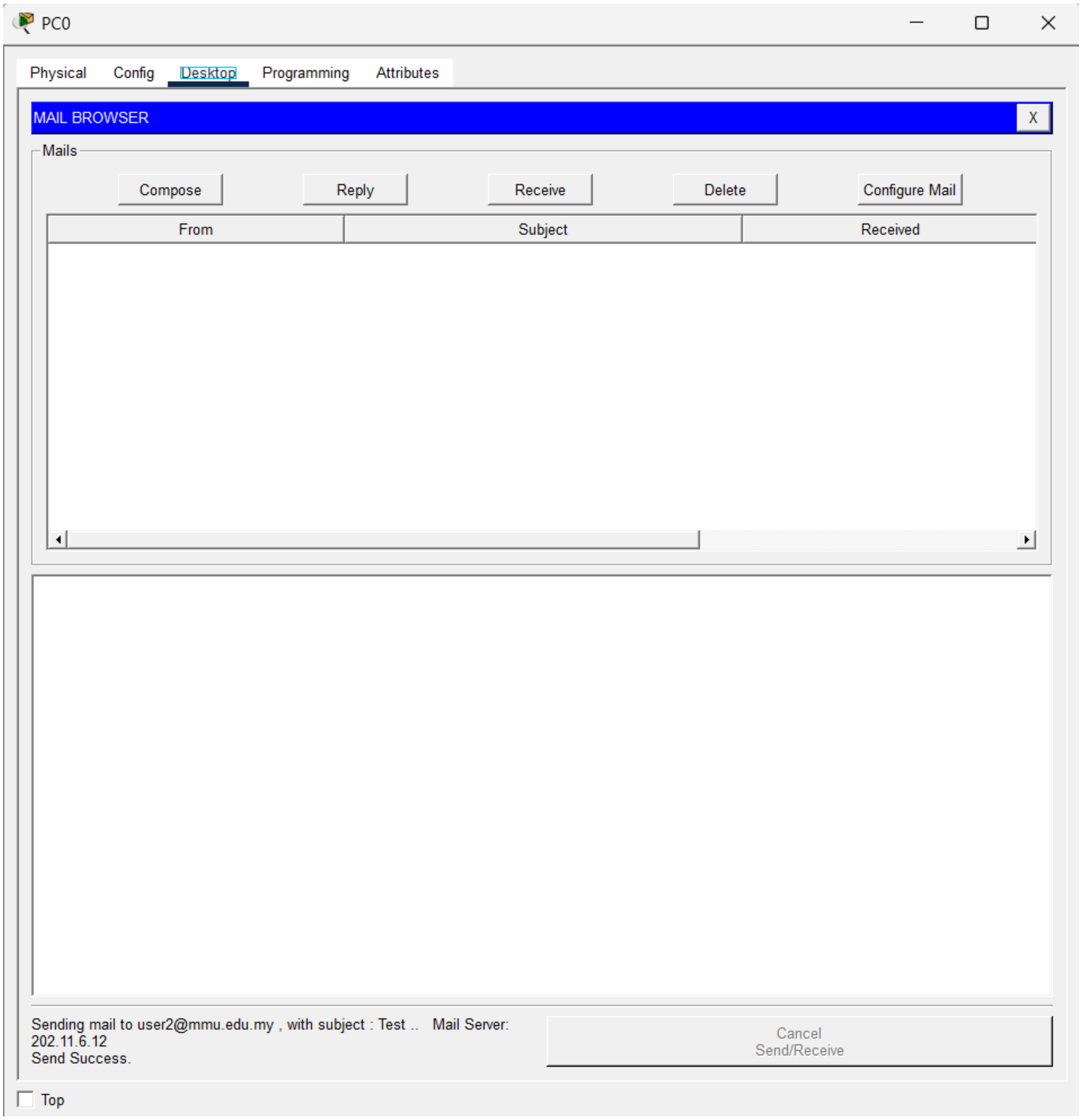
TEST 5 — FTP Service Test

```
C:\>ftp 202.11.6.13
Trying to connect...202.11.6.13
Connected to 202.11.6.13
220- Welcome to PT Ftp server
Username:cisco
331- Username ok, need password
Password:*****
230- Logged in
(passive mode On)
ftp>dir

Listing /ftp directory from 202.11.6.13:
0   : asa842-k8.bin                                      5571584
1   : asa923-k8.bin                                      30468096
2   : c1841-advipservicesk9-mz.124-15.T1.bin             33591768
3   : c1841-ipbase-mz.123-14.T7.bin                      13832032
4   : c1841-ipbasek9-mz.124-12.bin                       16599160
5   : c1900-universalk9-mz.SPA.155-3.M4a.bin             33591768
6   : c2600-advipservicesk9-mz.124-15.T1.bin             33591768
7   : c2600-i-mz.122-28.bin                              5571584
8   : c2600-ipbasek9-mz.124-8.bin                        13169700
9   : c2800nm-advipservicesk9-mz.124-15.T1.bin           50938004
10  : c2800nm-advipservicesk9-mz.151-4.M4.bin            33591768
11  : c2800nm-ipbase-mz.123-14.T7.bin                    5571584
12  : c2800nm-ipbasek9-mz.124-8.bin                      15522644
13  : c2900-universalk9-mz.SPA.155-3.M4a.bin             33591768
14  : c2950-i6q4l2-mz.121-22.EA4.bin                     3058048
15  : c2950-i6q4l2-mz.121-22.EA8.bin                     3117390
16  : c2960-lanbase-mz.122-25.FX.bin                     4414921
17  : c2960-lanbase-mz.122-25.SEE1.bin                   4670455
18  : c2960-lanbasek9-mz.150-2.SE4.bin                   4670455
19  : c3560-advipservicesk9-mz.122-37.SE1.bin            8662192
20  : c3560-advipservicesk9-mz.122-46.SE.bin             10713279
21  : c800-universalk9-mz.SPA.152-4.M4.bin               33591768
22  : c800-universalk9-mz.SPA.154-3.M6a.bin              83029236
23  : cat3k_caa-universalk9.16.03.02.SPA.bin             505532849
24  : cgr1000-universalk9-mz.SPA.154-2.CG                159487552
25  : cgr1000-universalk9-mz.SPA.156-3.CG                184530138
26  : homework.txt                                       22
27  : ie9k_iosxe.17.09.04.SPA.bin                        596133776
28  : ir800-universalk9-bundle.SPA.156-3.M.bin           160968869
29  : ir800-universalk9-mz.SPA.155-3.M                   61750062
30  : ir800-universalk9-mz.SPA.156-3.M                   63753767
31  : ir800_yocto-1.7.2.tar                              2877440
32  : ir800_yocto-1.7.2_python-2.7.3.tar                 6912000
33  : ir8340-mono-universalk9_iot.17.08.01a.SPA.pkg      685645824
34  : pt1000-i-mz.122-28.bin                             5571584
35  : pt3000-i6q4l2-mz.121-22.EA4.bin                    3117390
ftp>
```

Figure 5.5: Successful FTP login and directory access

TEST 6 — Email Send & Receive


PC0
Physical
Config
Desktop
Programming
Attributes
MAIL BROWSER
X
Mails
Compose
Reply
Receive
Delete
Configure Mail
From
Subject
Received
Sending mail to user2@mmu.edu.my , with subject : Test .. Mail Server: 202.11.6.12
Send Success.
Cancel
Send/Receive
Top

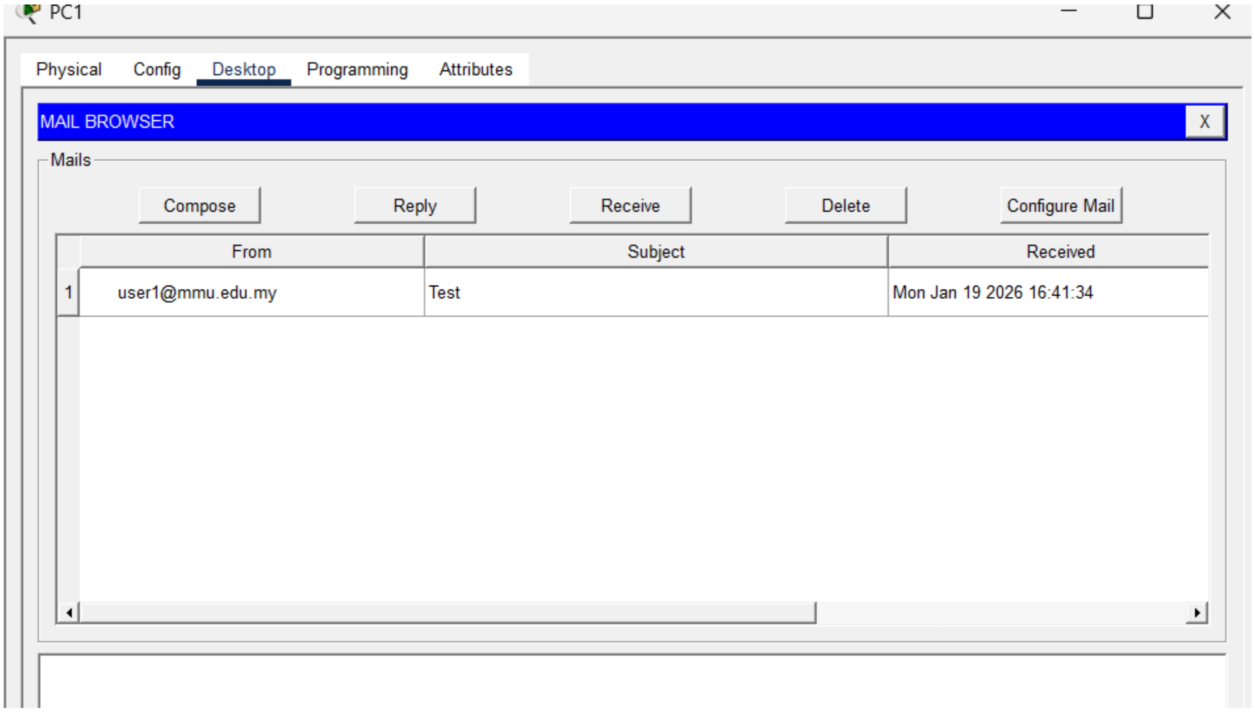


Figure 5.6: Successful email send and receive test between two wired clients using SMTP and POP3 services

5.1 Wireless + RADIUS Authentication

5.1.1 Wireless Network Overview

The wireless network was implemented using multiple access points connected to the core switch. Wireless clients obtain IPv4 addresses via DHCP and IPv6 addresses via SLAAC. Authentication is secured using a RADIUS server to ensure only authorised users can access the wireless network.

5.1.2 RADIUS Server Configuration

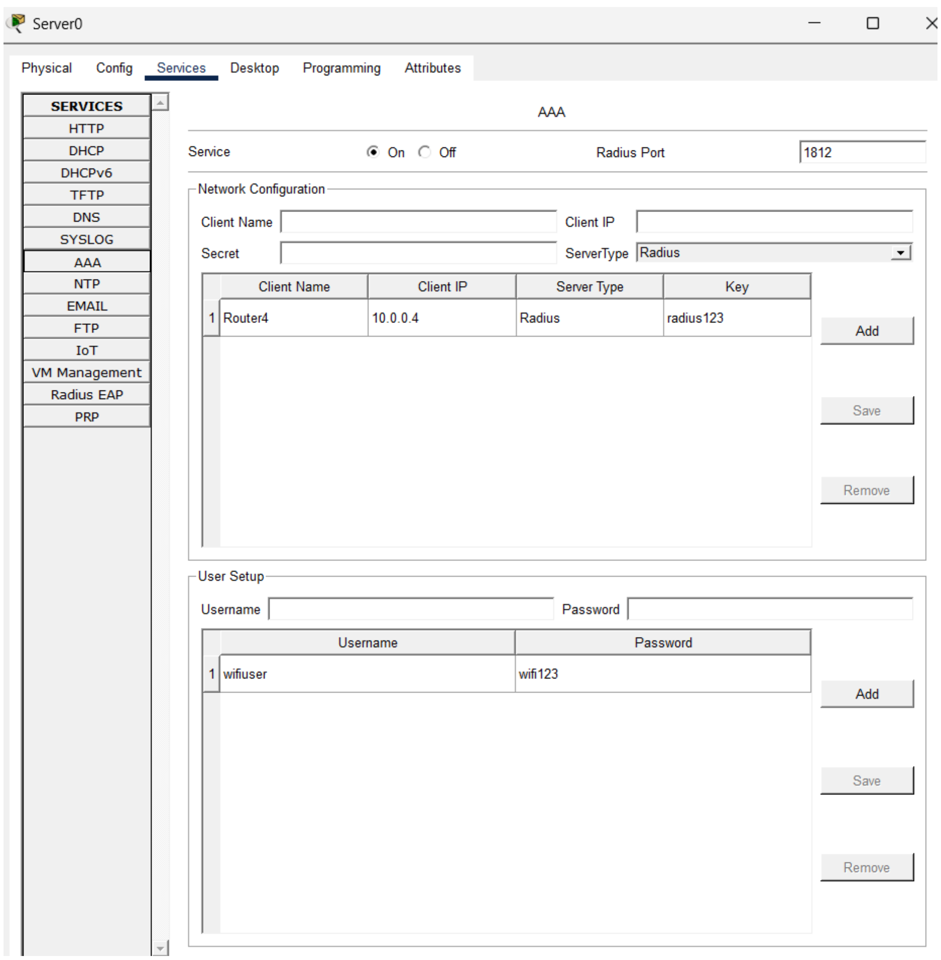


Figure 5.4: RADIUS server configuration showing authorised network client and wireless user credentials

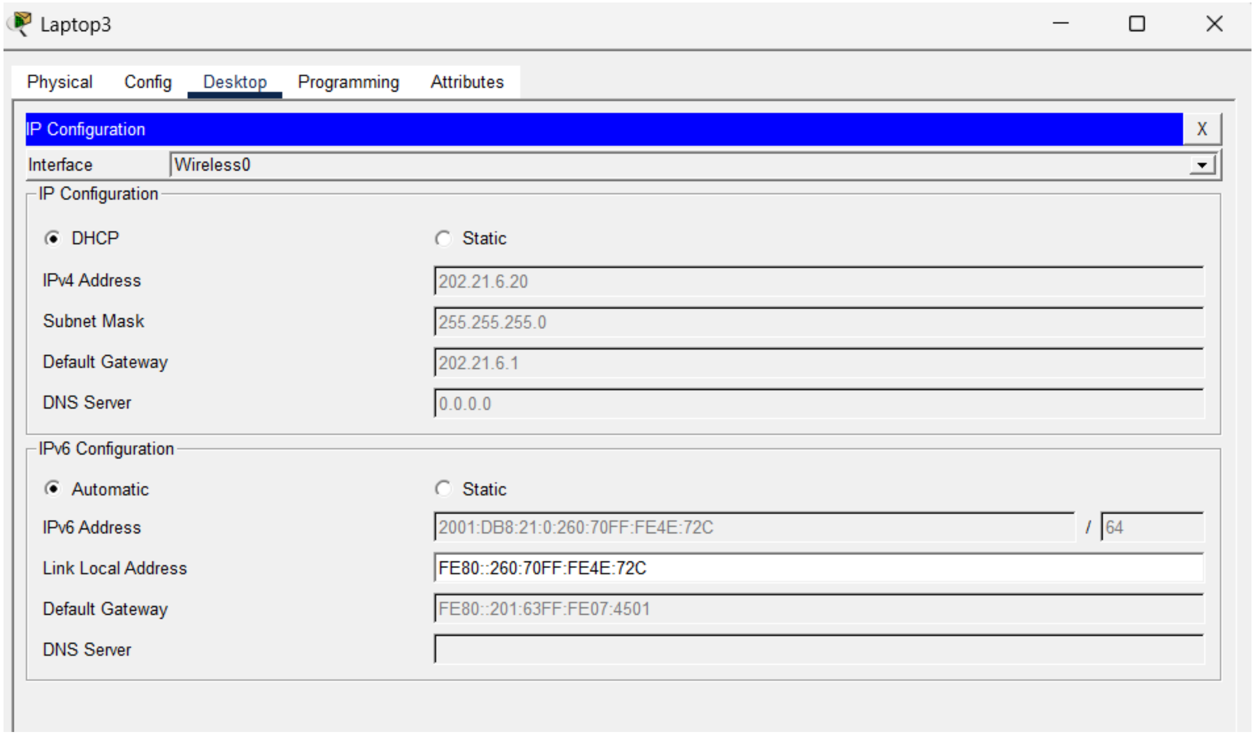


Figure 5.6: Wireless client successfully authenticated via RADIUS and assigned IP address through DHCP

6.1 Connectivity Tests

TEST 6.1.1 — IPv4 Ping Between Networks

Test 1 — Pink → Wireless

```
C:\>ping 202.11.6.1

Pinging 202.11.6.1 with 32 bytes of data:

Reply from 202.11.6.1: bytes=32 time<1ms TTL=254
Reply from 202.11.6.1: bytes=32 time<1ms TTL=254
Reply from 202.11.6.1: bytes=32 time<1ms TTL=254
Reply from 202.11.6.1: bytes=32 time<1ms TTL=254

Ping statistics for 202.11.6.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

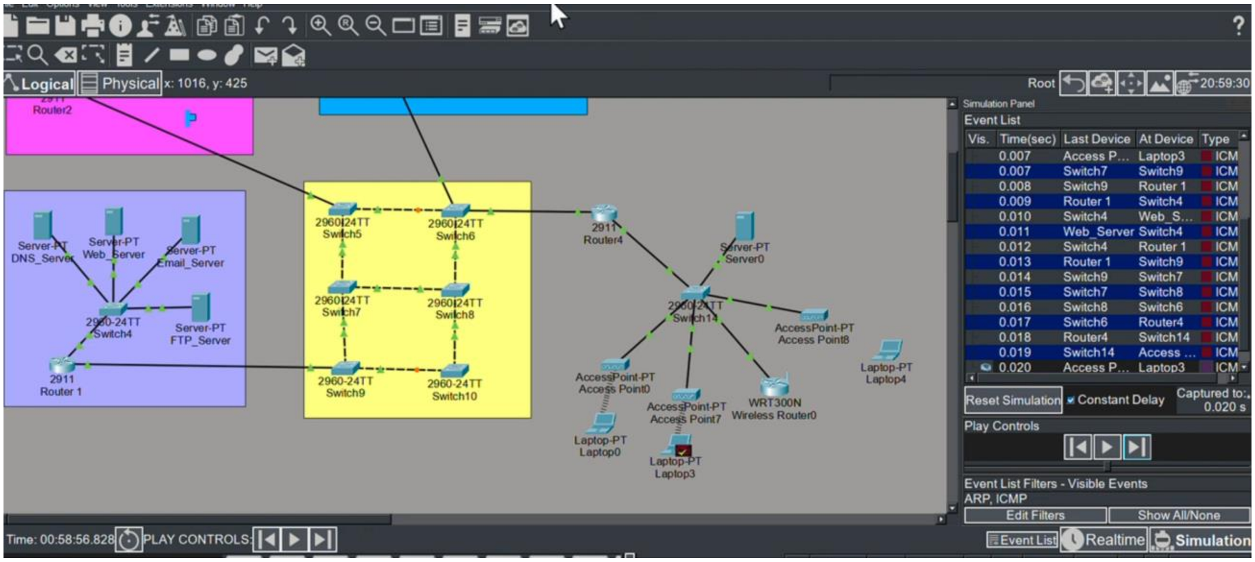
C:\>
```

```
Pinging 202.20.6.1 with 32 bytes of data:

Reply from 202.20.6.1: bytes=32 time=30ms TTL=254
Reply from 202.20.6.1: bytes=32 time=26ms TTL=254
Reply from 202.20.6.1: bytes=32 time=26ms TTL=254
Reply from 202.20.6.1: bytes=32 time=8ms TTL=254

Ping statistics for 202.20.6.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 8ms, Maximum = 30ms, Average = 22ms
```

IPv4 connectivity tests were conducted between multiple wired and wireless networks. Successful ICMP responses confirm that inter-network IPv4 routing is operational. Due to time constraints, selected representative paths were tested to validate network connectivity.



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	Laptop3	Web_Server	ICMP		0.000	N	0	(edit)	

Figure 6.7: Packet Tracer simulation mode illustrating successful ICMP packet flow across wired and wireless networks

YOUTUBE LINK:https://www.youtube.com/watch?v=BuJp4DuZnw

Group Marking Rubric (90%)

Assignment

CSN6224 Communication Networks

Trimester October/November 2025 Term 2530

Group Name:			Group ID:	20

This document contains the official assessment rubric for your assignment. Marks will be awarded based on the criteria and weightage specified in this rubric, which accounts for **90%** of the **total project marks**. All groups must follow this rubric when completing and presenting their work. The remaining **10%** of the marks will be based on the **Individual Self-Evaluation Form** submitted together with your final report.

Criteria	Weightage (%)	Score 4	Score 3	Score 2	Score 1	Score 0	Rating (Lecturer)
1. VLAN Segmentation & Switch Configuration	**15%**	VLANs, trunks, STP, redundancy fully correct & stable	Mostly correct with minor errors	VLANs correct but STP/trunks incomplete	Major errors in VLAN or trunk config	Not attempted	
2. IPv4 & IPv6 Addressing Plan (DOB-Based)	**10%**	Fully correct IPv4/IPv6 addressing with accurate tables	Mostly correct	Some errors or incomplete tables	Incorrect formatting or wrong addressing	Not attempted	
3. Dual-Stack Server Configuration	**10%**	All services (DNS, Web, Email, FTP) fully functional on IPv4 & IPv6	Most services work; minor issues	Some services functional	Only 1 service working / major issues	No servers configured	
4. Dynamic Routing (IPv4 + IPv6)	**10%**	Full OSPFv2/OSPFv3 (or RIP) convergence across all routers	Routing works but minor gaps	Partial routing; some unreachable networks	Incorrect routing protocols or major errors	No dynamic routing	
5. Wireless + RADIUS Authentication	**10%**	All 3 WLANs functional with stable WPA2-Enterprise, DHCP & RADIUS	RADIUS mostly works with minor issues	WLAN OK, RADIUS partially working	Incorrect WLAN/RADIUS setup	Not attempted	
6. End-to-End Connectivity & Simulation Testing	**10%**	All IPv4/IPv6 ping, services, RADIUS login, simulation verified	Most tests successful	Partial connectivity	Few tests working	No testing conducted	
7. Report Quality & Documentation	**10%**	Professional, clear, complete, with appendices & evidence	Well-written and complete	Adequate but lacks depth or clarity	Poorly organized	No report submitted	
8. Presentation (Slides + Video)	**10%**	Excellent clarity, structure, teamwork, technical accuracy	Good delivery with minor gaps	Basic presentation	Poor clarity or missing parts	No presentation submitted	

9. Bonus: Enhancements (SSH, ACLs, Redundancy, Syslog/SNMP)	**5%**	3+ enhancements fully functional	Multiple enhancements with minor issues	1–2 enhancements partially working	Attempted but incorrect	No enhancements	

Weekly Project Guidelines / Checklist (Week 5 – 13) Assignment

CSN6224 Communication Networks

Trimester October/November 2025 Term 2530

Group Name:	Boys		Group ID:	20

These guidelines serve as a weekly reminder of the suggested tasks for your group. They are intended to help you plan and complete your project effectively. You are not required to follow them strictly, and your group may adjust the schedule as needed based on progress and workload.

Week	Task Title	Task Item	Checklist
Week 5	Group Formation	Form group & confirm member list	DONE
		Set up communication channel	DONE
		Assign roles (Leader, Technical Lead, Documentation Lead, Presentation Lead, QA)	DONE
		Conduct first meeting	DONE
Week 6	Requirements Analysis	Study assignment requirements	DONE
		Develop project scenario	DONE
		Divide tasks among group members	DONE
		Prepare high-level network design	DONE
		Lecturer check-in: confirm scope and design idea	DONE
Week 7	Initial Network Design	Draft topology in Packet Tracer	DONE
		Start IP addressing plan	DONE
		Identify required protocols	DONE
		Lecturer check-in: verify topology and addressing draft	DONE
Week 8	Requirements Mapping	Finalize topology	DONE
		Begin basic configurations	DONE
		Lecturer check-in: confirm all requirements covered	DONE
Week 9	Core Implementation	Configure routing, switching, DHCP/NAT/ACL	DONE

		Conduct basic connectivity tests	DONE
		Lecturer check-in: demonstrate partial working network	DONE
Week 10	Troubleshooting & Enhancement	Fix issues and refine configurations	DONE
		Add enhancements (if allowed)	DONE
		Lecturer check-in: verify functionality	DONE
Week 11	Documentation & Review	Finalize topology, tables, configuration summaries, test results	DONE
		QA check	DONE
		Lecturer check-in: review near-final work	DONE
Week 12	Final Report & Video	Finalize written report	DONE
		Finalize presentation slides	DONE
		Record project video presentation	DONE
		Upload video to YouTube (Unlisted)	DONE
Week 13	Submission to eBwise	Submit Packet Tracer file (.pkt)	DONE
		Submit PDF report	DONE
		Submit presentation slides (.pptx or other format)	DONE

Group Name: GROUP 20

NO.	TASK DESCRIPTION	MEMBER-IN-CHARGE	DATE ASSIGNED	DATE COMPLETED	REMARK
1	Network topology design & overall planning	Raamesh (Leader)	28/12/2026	04/01/2026	Initial network design
2	IPv4 addressing plan & subnetting	Raamesh	10/01/2026	23/01/2026	DOB-based IPv4 addressing
3	IPv6 addressing plan (SLAAC & static)	Raamesh, Tharunnesh	10/01/2026	28/01/2026	IPv6-only & dual-stack setup
4	VLAN segmentation & trunk configuration	Tharunnesh	10/01/2026	29/01/2026	VLANs & trunk links
5	Core network redundancy & STP configuration	Raamesh	10/01/2026	29/01/2026	STP enabled
6	Dynamic routing (OSPFv2 & OSPFv3)	Raamesh, Hareen	25/01/2026	29/01/2026	IPv4 & IPv6 routing
7	Dual-stack server configuration (DNS, HTTP, FTP, Email)	Raamesh	25/01/2026	29/01/2026	All services configured
8	Wireless network &	Hareen	25/01/2026	29/01/2026	Wireless authentication

	RADIUS (AAA) setup				
9	Connectivity testing & simulation mode verification	Raamesh	25/01/2026	29/01/2026	IPv4, IPv6 & services tested
10	Report formatting, screenshots & final review	All Members	25/01/2026	29/01/2026	Final submission prep

**Note: You may edit the format of this page.*

Individual Self-Evaluation Form (10%)
Assignment

CSN6224 Communication Networks
Trimester October/November 2025 Term 2530

Student Name:	RAAMESH A/L MANIRAJAN		Student ID:	243UC247NZ
Group Name:	GROUP 20		Group ID:	20

Each group member must complete and sign this form to verify the accuracy of their self-evaluation. Please **attach** the completed form together with your group report during submission.

If you have **concerns** or **disagree** with any part of a group member's self-evaluation, please **email** the lecturer **privately**. The lecturer will review the matter, and appropriate action will be taken if the concern is substantiated.

Criteria	Weightage (%)	4 – Excellent	3 – Good	2 – Fair	1 – Poor	0 – None	Your Rating (Number)
Attendance	2	Attended all meetings except excused official tasks (with proof).	Missed 1 meeting with acceptable reason.	Missed 2–3 meetings; explanation acceptable but weak.	Frequently absent; weak or unclear reasons.	No attendance / no explanation.	4
Task Completion	3	Completed all tasks on time with clear evidence.	Completed most tasks; minor delays.	Completed some tasks; multiple delays.	Very minimal task completion.	No completed tasks.	4
Communication	1.5	Always responsive;	Mostly responsive; occasional delay.	Sometimes responds; inconsistent.	Rarely responds; slow communication.	No communication at all.	4

		active in chat & group activities.					
Contribution	2	Actively contributes ideas and solutions in discussions.	Regular contributions; useful input.	Occasional contributions; basic input.	Rarely contributes; passive.	No contribution.	4
Initiative	1.5	Proactively initiates tasks and helps others consistently.	Often shows initiative and helps when needed.	Sometimes helps; limited initiative.	Rarely initiates; only helps when asked.	No initiative or avoids helping.	4

By signing below, you confirm that all information provided in this form is true and accurate.

Declaration

✓		I hereby confirm that the above self-evaluation is accurate.

RAAMESH

………………………………………………………….

Student Name: RAAMESH A/L MANIRAJAN

Student ID: 243UC247NZ

Date:30/1/2026

Individual Self-Evaluation Form (10%)
Assignment
CSN6224 Communication Networks
Trimester October/November 2025 Term 2530

Student Name:	THARUNNESH MAHALINGA MOORTHY		Student ID:	241UC240NP
Group Name:	GROUP 20		Group ID:	20

Each group member must complete and sign this form to verify the accuracy of their self-evaluation. Please **attach** the completed form together with your group report during submission.

If you have **concerns** or **disagree** with any part of a group member's self-evaluation, please **email** the lecturer **privately**. The lecturer will review the matter, and appropriate action will be taken if the concern is substantiated.

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Declaration

✓		I hereby confirm that the above self-evaluation is accurate.

THARU

…………………………………………………………..

Student Name: THARUNNESH MAHALINGA MOORTHY

Student ID: 241UC240NP

Date:30/1/2026

Individual Self-Evaluation Form (10%)
Assignment

CSN6224 Communication Networks
Trimester October/November 2025 Term 2530

Student Name:	HAREEN A/L VIJAYA KUMAR		Student ID:	242UC244Q0
Group Name:	GROUP 20		Group ID:	20

Each group member must complete and sign this form to verify the accuracy of their self-evaluation. Please **attach** the completed form together with your group report during submission.

If you have **concerns** or **disagree** with any part of a group member's self-evaluation, please **email** the lecturer **privately**. The lecturer will review the matter, and appropriate action will be taken if the concern is substantiated.

Criteria	Weightage (%)	4 – Excellent	3 – Good	2 – Fair	1 – Poor	0 – None	Your Rating (Number)
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Declaration

✓		I hereby confirm that the above self-evaluation is accurate.

HAREEN

………………………………………………………

Student Name: HAREEN A/L VIJAYA KUMAR

Student ID: 242UC244Q0

Date:30/1/2026